

I/O OPERATIONS

Program Controlled

In Program Control I/O the data transfers are completely under the control of the microprocessor. i.e. an I/O operation takes place only when an I/O transfer instruction is executed.

In some situation microprocessor has to test the readiness of the I/O device before performing the data transfer. For this purpose it has to continuously check the readiness by reading the device's status register.

Checking the readiness is an addition I/O operation

Interrupt Driven I/O

In these I/O operations a device signals the microprocessor about its readiness by generating an interrupt signal.

The microprocessor doesn't have to spend time to check the device continuously.

Once the device has informed the microprocessor about its readiness the data transfer take place under microprocessor control.

The microprocessor saves its context upon receiving the interrupt so that it could resume program execution from that same point where it was interrupted.

DMA

The Data transfer directly takes place between an I/O device and the memory.

The microprocessor sets up the data transfer by sending necessary information to the DMA device.

After that DMA device / controller request and control the data transfer.

Device Selection Topic to be discussed in Class

Interrupt

Draw Attention of Microprocessor

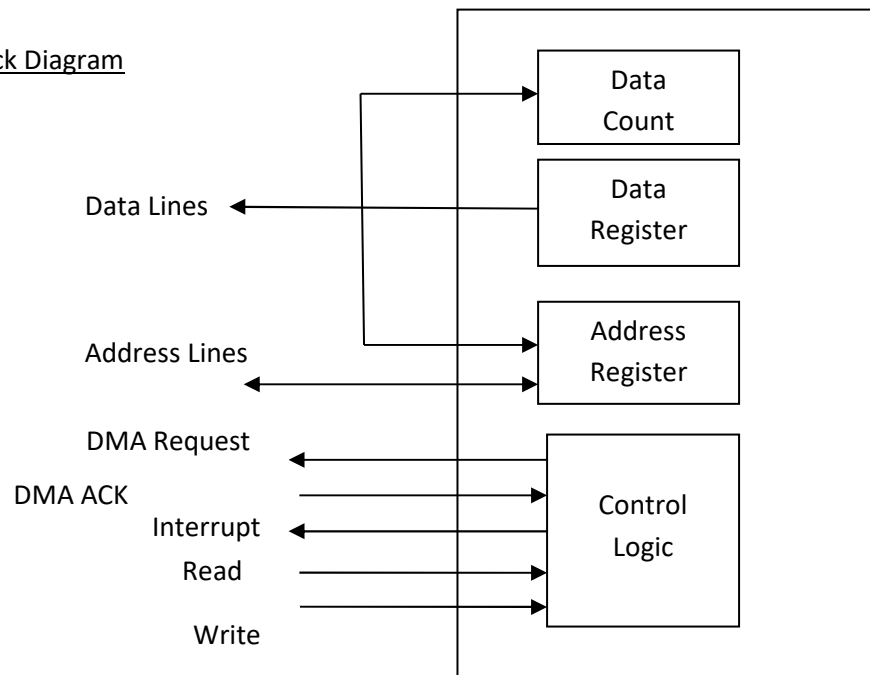
- Essentially an interrupt is a subroutine call initiated by the external hardware.
- This subroutine is executed in response to the Interrupt Request (IR) of the device.
- When an I/O Device requires service it sets its IR flip flop. This is connected to the INTR pin of the microprocessor.
- This flip flop strobes the I/O device's interrupt it is acknowledge by the microprocessor.
- The IR (Interrupt Request) may occur at any point during program execution.
- Upon the occurrence of an interrupt the execution of the current instruction is completed and then the control is transferred to the Interrupt Service Routine (ISR).
- In order to allow the resumption of the interrupted program, when the ISR is finished, the Program Counter is automatically saved before control is transferred to the ISR.
- In addition to servicing the Interrupt Request (IR) the subroutine does two important things:
 - In beginning, it saves the contents of any register it uses, on the stack.
 - In the end, before returning the control to the interrupted program, it RESTOREs the contents of the Registers, it saved in the beginning.

Drawbacks of Programmed and Interrupt Driven I/O:

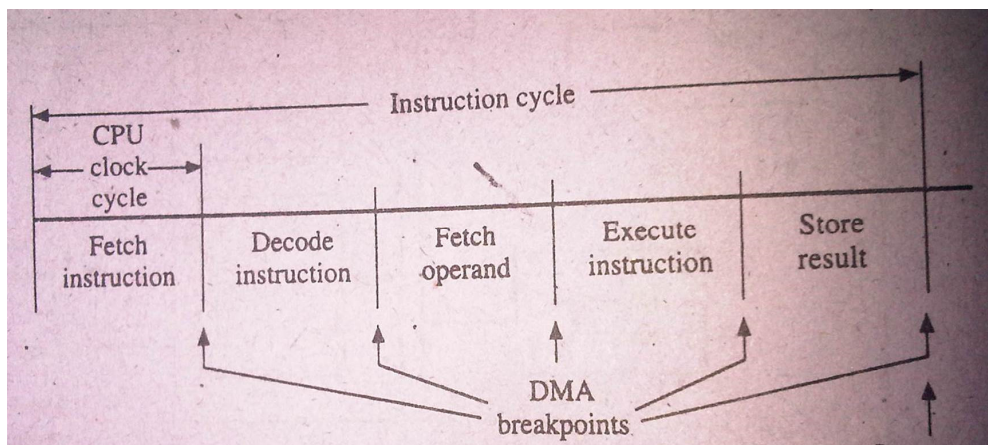
- The I/O transfer rate is limited by the speed with which processor can test and service a device.
- The Processor is tied up in managing an I/O transfer, a number of instructions must be executed for each I/O transfer.

DMA (Direct Memory Access)

Typical DMA Block Diagram



- DMA involves an additional module on the system bus (The DMA Module)
- Special control lines connect the IO devices to CPU to send DMA Request.
- DMA request causes the CPU to suspend its current activities, at appropriate BREAKPOINTS and attend the DMA Request.
- A DMA Request requires the CPU to grant control of system bus to requesting device.
- CPU take control of the bus at end of transfer.

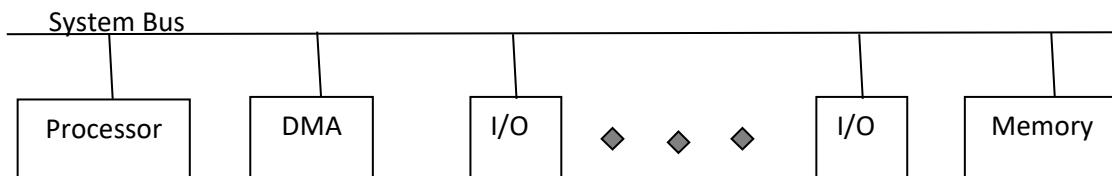


- A DMA Module contain Data Buffer Register, Address Register and Data Count Register.
- These registers enable data transfer to or from a contiguous region of memory.
- Data can be transferred in several different ways (DMA modes)
- In a DMA Block Transfer a sequence of data word of specified length (Data Count) is transferred in a single burst, DMA controller (DMA Module) is master of memory bus.

- An Alternative technique is Cycle Stealing, the DMA module use system bus to transfer one data word, after which bus control is returned to CPU. Long blocks of IO data is transferred by a sequence of DMA bus transactions intercepted with CPU bus transactions.
- A DMA interface can be designed, so that bus cycles are stolen only when CPU is not actually using the system bus. This mode is called Transparent DMA.
- DMA transfer proceeds as follows:
 - o CPU loads Data Count (DC) and Address Register (AR). AR contains base address of memory region to be used for data transfer. DC contains number of words to be transferred from that region.
 - o When DMA is ready to transmit / receive data, it issues DMA request. CPU waits for the next DMA Breakpoint, at breakpoint it relinquish control of system bus and issues DMA Acknowledgement.
 - o DMA controller now transfers data to or from main memory. After one word is transferred AR is incremented and DC is decremented.
 - o If DC is not zero and I/O device is not ready to send / receive data, the system bus is released in favor of CPU and DMA request line deactivated.
 - o If DC is zero, DMA relinquish control of system bus, signal CPU with interrupt that DMA operation completes.

Alternative DMA Configurations

Single Bus, Detached DMA



I/O Bus Configuration

